DISCRETE MATH · ELECTION THEORY

Voting, Power & Apportionment Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Voting, Power & Apportionment Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Schedule: 14 A > B > C, 10 C > B > A, 8 B > C > A, 3 C > A > B. Predict the plurality winner. Is it a majority?
   A with 14 of 35 votes; not a majority (18 needed).
2. In [6: 4, 3, 2], who do you think has the most power? Is it proportional to weight?
   P1, but not proportionally: P1 is critical 3 times, P2 and P3 once each (Banzhaf 3/5, 1/5, 1/5).

2 Explore use the tool

1. Enter the schedule from Predict 1. Record the winner under each method.
   Plurality A (14); Borda B (A 66, B 75, C 69); plurality with elimination C (drop B, C 21–14); pairwise B.
2. Is there a Condorcet winner? Which methods violate the Condorcet criterion here?
   B beats A 18–17 and C 22–13. Plurality (A) and elimination (C) violate it.
3. Weighted voting [6: 4, 3, 2]: list the winning coalitions and critical players.
   {P1, P2} both critical; {P1, P3} both critical; {P1, P2, P3} only P1 critical.
4. Apportionment: 130, 690, 410 with Hamilton. Record seats for 13 and 14 seats. What happens?
   13 seats: 2, 7, 4; 14 seats: 1, 8, 5. State 1 loses a seat: the Alabama paradox.

3 Record use the tool and fill in the table

Weighted voting systems

System	Dictator?	Veto power	Dummy	Banzhaf counts
[6: 4, 3, 2]	no	P1	none	3, 1, 1
[4: 3, 2, 2]	no	none	none	2, 2, 2
[6: 4, 2, 1]	no	P1, P2	P3	2, 2, 0
[5: 5, 2, 1]	P1	P1	P2, P3	4, 0, 0

Apportion 10 seats: populations 450, 330, 220 (SD = 100)

Method	Divisor used	Seats
Hamilton	SD = 100	5, 3, 2
Jefferson	any d with 82.5 < d ≤ 90 (for example 90)	5, 3, 2
Adams	any d with 110 ≤ d < 112.5 (for example 111)	5, 3, 2

4 Explain what did you discover?

1. A majority is more than half of the votes.
2. A player with weight at least the quota is a dictator; a player who is never critical is a dummy.
3. Hamilton always satisfies the quota rule but can produce the Alabama paradox; divisor methods avoid the paradox but can violate the quota rule.
4. Use your Explore 1 results to explain why "the winner" of an election depends on the method, and why the method should be chosen before the votes are counted.
   The same ballots gave three different winners (A, B, C) under four reasonable methods, so the choice of method can decide the outcome; choosing after seeing the ballots would let someone pick the method that favors their candidate.

5 Apply a new situation, no tool

1. [10: 6, 5, 4]: find the Banzhaf power index of each player.
   Winning: {P1, P2}, {P1, P3}, {P1, P2, P3} ({P2, P3} has only 9). Critical counts 3, 1, 1, so 3/5, 1/5, 1/5.
2. Is [7: 4, 3, 2, 1] a valid system? Does anyone have veto power?
   Valid (total 10, 7 > 5). P1 has veto power (others total 6 < 7).
3. Populations 5100, 1600, 1500, 1200; 20 seats; Jefferson. Find a divisor and the seats. Is the quota rule met?
   SD = 470; d = 425 gives 12, 3, 3, 2. State 1's quota is 10.85, so 12 violates the quota rule.
4. Ann bids $300 for a house and $120 for a car; Ben bids $240 and $180 (sealed bids). Who gets what, and what cash changes hands?
   Ann gets the house, Ben the car; fair shares $210 each; after splitting the $60 surplus Ann pays Ben $60.

Teacher notes

Intended discovery: Reasonable voting methods can disagree; fairness criteria can each be violated; power depends on critical appearances rather than weight; Hamilton satisfies quota but has paradoxes while divisor methods avoid paradoxes but can violate quota.

Likely misconceptions:

- Treating plurality as majority.
- Assuming power is proportional to weight.
- Rounding quotas normally and forgetting that seats must add up.

Answers that can vary: Divisors that work form an interval, and any value in it is correct (Jefferson: $82.5 < d \leq 90$; Adams: $110 \leq d < 112.5$; Apply 3: any d with $400 < d \leq 425$).